

AIDS1- PT2 Question Bank

TE I.T.Sem 6 (Jan-May 2025)

1. Illustrate with diagram how goal based agent works.
2. What is learning agent? Explain with diagram.
3. Explain utility based agent architecture with diagram.
4. What are different environment types? Give example of each.
5. Describe PEAS and also write down the PEAS representations for
 - a. Medical diagnosis system.
 - b. Automated Taxi Driver
6. Describe different uninformed search techniques and compare them based on their time complexity, space complexity, completeness and optimality.
7. Formulate the 8 queen problem
8. What are heuristic functions? Where are they used? Give heuristic functions for 8 puzzle problem.
9. What is Unification? Give example.
10. What is Skolemization? Explain Skolem constant and Skolem function.
11. Explain uniform cost search and best first search in detail with examples and compare.
12. What is Bidirectional Search? Comment about completeness, optimality, time and space complexity.
13. Differentiate between forward and backward chaining.
14. Numericals on Resolution by refutation in First order Logic.
15. State the rules to convert First Order Logic statements in CNF. Explain each rule with example.
16. Numerical on Conversion of given statements to First order logic.
17. State A* algorithm and prove that it is admissible.
18. Numerical on water-jug and crypto-arithmetic problems.

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